

Riemannian Geometry

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1 Smooth Manifolds

1.1 Simple pre-requisites

This chapter will be built heavily on the basics of topology. This section will be a recap of some important properties.

Definition 1.1 (A topology on a set). Given a set X , we define τ to be a collection of subsets from X that follow certain rules:

1. The empty set $\emptyset$ and X are in τ .
2. If we have a collection of sets in τ , then their union is also in τ . i.e. If $U_\alpha \in \tau \forall \alpha \in I$, then $\bigcup_{\alpha \in I} U_\alpha \in \tau$
3. If $U_1, \dots, U_n \in \tau$ then $\bigcap_{i=1}^n U_i \in \tau$

We call the elements in τ open sets and we use this along with our set X to create a topological space.

Definition 1.2 (A topological space). A topological space is the pair (X, τ_X) where X is just a set and τ_X is the topology on X

Instead of always writing (X, τ_X) , we use X to refer to the space.

Definition 1.3 (A continuous function between topological spaces). Given 2 topological spaces X, Y , a function $f : X \rightarrow Y$ is said to be continuous if for every open set $V \subseteq Y$, the preimage $f^{-1}(V) = \{x \in X \mid f(x) \in V\}$ is open in X .

This definition does not depend on a metric, so it allows us to define continuity on general topological spaces.

Definition 1.4 (Homeomorphisms). Given 2 topological spaces X and Y . A function $f : X \rightarrow Y$ is called a homeomorphism if:

1. f is bijective
2. f is continuous
3. f^{-1} is continuous

If 2 spaces X and Y have a homeomorphism between them, we call them homeomorphic and use $X \cong Y$ to represent that.

This notion of topology allows us to define spaces that locally resemble Euclidean space without requiring a global coordinate system.

1.2 Topological Manifolds

Definition 1.5 (Topological Manifold). Let M be a topological space. M is a **topological n -manifold** (topological manifold of dimension n) if it is:

- A Hausdorff space: For any 2 distinct points $p, q \in M$, there exists 2 disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- Second countable: there exists a countable basis for the topology of M
- M is locally Euclidean of dimension n : each point $p \in M$ has a neighbourhood that is homeomorphic to an open subset of $\mathbb{R}^n$

A very simple example of a topological manifold is the space $\mathbb{R}^n$ itself. We'll cover more interesting spaces later.

Definition 1.6 (Coordinate Charts). Let M be a topological n -manifold. A coordinate chart on M is a pair (U, φ) where U is an open subset of M and φ is a homeomorphism $f : U \rightarrow \hat{U}$ where $\hat{U} \subseteq \mathbb{R}^n$. We say the chart is centred at p if $\varphi(p) = 0$

Charts are very useful, as without them, calculus on these manifolds would be next to impossible. Coordinate charts give us a way of easily moving from the complex, strange manifolds to simple Euclidean space. Now that we have these definitions, we can start giving examples of more interesting manifolds.

Example 1.1 (Spheres). For any $n \geq 0$, the n -sphere is defined as:

$$\mathbb{S}^n = \left\{ (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 = 1 \right\}$$

If we equip this from the subspace topology inherited from $\mathbb{R}^{n+1}$, This forms a topological n -manifold. Since $\mathbb{R}^{n+1}$ is a metric space, it is Hausdorff and second countable. Because $\mathbb{S}^n$ uses this subspace topology, it also inherits them.

To show that it is locally Euclidean, we must show that every point $p \in \mathbb{S}^n$ has a neighbourhood that is part of a coordinate chart (U, φ) .

Let $N = (0, \dots, 0, 1)$ denote the North pole and $S = (0, \dots, 0, -1)$ denote the south pole. We define the 2 open sets of $\mathbb{S}^n$ as:

$$U_N = \mathbb{S}^n \setminus \{N\}, \quad U_S = \mathbb{S}^n \setminus \{S\}$$

First consider $p \in \mathbb{S}^n$ such that $p \neq N$. Then $p \in U_N$. We can now define the stereographic projection map $\varphi_N : U_N \rightarrow \mathbb{R}^n$ as:

$$\varphi_N(x_1, \dots, x_{n+1}) = \left(\frac{x_1}{1 - x_{n+1}}, \dots, \frac{x_n}{1 - x_{n+1}} \right)$$

This is clearly a homeomorphism as it is bijective and continuous. Its inverse is also continuous; however, due to limitations (i'm lazy), it will not be shown

in this document. Since φ_N is a homeomorphism between U_N and $\mathbb{R}^n$, the pair (U_N, φ_N) is a coordinate chart containing p . Now consider the point $p = N$. This would mean $p \in U_S$. We can define the projection from the south pole as:

$$\varphi_S(x_1, \dots, x_{n+1}) = \left(\frac{x_1}{1 + x_{n+1}}, \dots, \frac{x_n}{1 + x_{n+1}} \right)$$

Since it is very similar to φ_N , we can use similar reasoning/proof methods to show it is also a homeomorphism. It also means that (U_S, φ_S) is also a valid coordinate chart containing N . Since every point is contained in at least one of these charts, $\mathbb{S}^n$ is locally Euclidean of dimension n . We have now officially shown that $\mathbb{S}^n$ is a topological n -manifold.

Example 1.2 (Product Manifolds). Suppose $M_1, \dots, M_k$ are topological manifolds of dimensions $n_1, \dots, n_k$ respectively. The product space $M_1 \times \dots \times M_k$ is a topological manifold of dimension $n_1 + \dots + n_k$ as follows. It is obviously Hausdorff and second countable, so the only property that needs to be checked is locally Euclidean. Given any point $(p_1, \dots, p_k) \in M_1 \times \dots \times M_k$, we can choose the coordinate chart (U_i, φ_i) for each M_i with $p_i \in U_i$. The product map:

$$\varphi_1 \times \dots \times \varphi_k : U_1 \times \dots \times U_k \rightarrow \mathbb{R}^{n_1 + \dots + n_k}$$

is a homeomorphism onto its image, which is just a product open subset of $\mathbb{R}^{n_1 + \dots + n_k}$. Thus $M_1 \times \dots \times M_k$ is a topological manifold of dimension $n_1 + \dots + n_k$.

Example 1.3 (Tori). for $n \in \mathbb{N}$, the n -torus is the product space: $\mathbb{T}^n = \mathbb{S}_1^1 \times \dots \times \mathbb{S}_n^1$. Since we proved that product manifolds are manifolds, it follows pretty simply that the n -torus is a manifold as well.

Before we finish this section, one thing we need to introduce is the notion of an Atlas.

Definition 1.7 (Atlas). Given a manifold M , an atlas is a collection of coordinate charts $\mathcal{A} = \{(U_\alpha, \varphi_\alpha) : \alpha \in I\}$ such that: $M = \bigcup_{\alpha \in I} U_\alpha$

1.3 Smooth Structures

Topological manifolds are useful for studying topological properties, but they break down when introducing calculus. The problem is that derivatives are not homeomorphism-invariant: the map $\varphi(u, v) = (u^{1/3}, v^{1/3})$ is a homeomorphism $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, yet $f(x, y) = x$ is smooth while $f \circ \varphi$ is not differentiable at the origin. To do calculus on manifolds, we need more structure.

Definition 1.8 (Smooth map and diffeomorphism). Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open. A map $f : U \rightarrow V$ is **smooth** (or C^∞) if all its component functions have continuous partial derivatives of all orders. If f is additionally bijective with a smooth inverse, it is called a **diffeomorphism**.

Definition 1.9 (Transition map). Let (U, φ) and (V, ψ) be two charts on a topological manifold M with $U \cap V \neq \emptyset$. The **transition map** from φ to ψ is the composite:

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \rightarrow \psi(U \cap V)$$

This is a map between open subsets of $\mathbb{R}^n$, so smoothness is defined in the ordinary sense.

Definition 1.10 (Smoothly compatible charts). Two charts (U, φ) and (V, ψ) are **smoothly compatible** if either $U \cap V = \emptyset$ or the transition map $\psi \circ \varphi^{-1}$ is a diffeomorphism.

Note: to verify smooth compatibility, it is enough to check that $\psi \circ \varphi^{-1}$ is smooth and injective with nonsingular Jacobian at each point; the inverse is then automatically smooth.

Definition 1.11 (Smooth atlas). An atlas $\mathcal{A}$ on M is a **smooth atlas** if any two charts in $\mathcal{A}$ are smoothly compatible. Since $\varphi \circ \psi^{-1} = (\psi \circ \varphi^{-1})^{-1}$ is also a transition map, we only need to check smoothness in one direction.

The issue is that many different smooth atlases can determine the same collection of smooth functions on M — they are morally the same smooth structure. We resolve this by passing to a maximal atlas.

Definition 1.12 (Smooth structure). A smooth atlas $\mathcal{A}$ on M is **maximal** if it is not properly contained in any larger smooth atlas; equivalently, any chart smoothly compatible with every chart in $\mathcal{A}$ is already in $\mathcal{A}$. A **smooth structure** on M is a maximal smooth atlas.

Definition 1.13 (Smooth manifold). A **smooth manifold** is a pair $(M, \mathcal{A})$ where M is a topological manifold and $\mathcal{A}$ is a smooth structure on M .

In practice, we never need to write down a maximal atlas explicitly. The following proposition tells us we only need to specify any smooth atlas.

Lemma 1.1 (Proposition 1.17). Let M be a topological manifold.

1. Every smooth atlas $\mathcal{A}$ for M is contained in a unique maximal smooth atlas, called the smooth structure determined by $\mathcal{A}$.
2. Two smooth atlases determine the same smooth structure if and only if their union is a smooth atlas.

Proof sketch of (1). Let $\bar{\mathcal{A}}$ be the set of all charts smoothly compatible with every chart in $\mathcal{A}$. For any two charts $(U, \varphi), (V, \psi) \in \bar{\mathcal{A}}$ and any $p \in U \cap V$, pick a chart $(W, \xi) \in \mathcal{A}$ with $p \in W$. Then $\psi \circ \varphi^{-1} = (\psi \circ \xi^{-1}) \circ (\xi \circ \varphi^{-1})$ is smooth near $\varphi(p)$ as a composition of smooth maps, so $\bar{\mathcal{A}}$ is itself a smooth atlas. Maximality and uniqueness follow readily. $\square$

Example 1.4 ($\mathbb{R}^n$). The single chart $(\mathbb{R}^n, \text{Id}_{\mathbb{R}^n})$ is trivially a smooth atlas. The smooth structure it determines is called the **standard smooth structure** on $\mathbb{R}^n$.

Example 1.5 (A different smooth structure on $\mathbb{R}$). The map $\psi(x) = x^3$ is a homeomorphism $\mathbb{R} \rightarrow \mathbb{R}$. The atlas $\{(\mathbb{R}, \psi)\}$ defines a smooth structure on $\mathbb{R}$. This is not the standard one: the transition map $\text{Id} \circ \psi^{-1}(y) = y^{1/3}$ is not smooth at 0, so these two charts are not smoothly compatible.

Example 1.6 (Open submanifolds). If M is a smooth n -manifold and $U \subseteq M$ is open, then U is itself a smooth n -manifold: take the atlas $\{(V, \varphi|_V) : (V, \varphi) \text{ smooth chart for } M, V \subseteq U\}$.

Example 1.7 ($\mathbb{S}^n$). The graph coordinate charts $\{(U_i^\pm, \varphi_i^\pm)\}$ from Example 1.1 form a smooth atlas for $\mathbb{S}^n$: one can compute the transition maps explicitly and verify they are diffeomorphisms. This defines the **standard smooth structure** on $\mathbb{S}^n$.

Example 1.8 ($GL(n, \mathbb{R})$). The general linear group $GL(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) : \det A \neq 0\}$ is an open subset of $M(n, \mathbb{R}) \cong \mathbb{R}^{n^2}$, so it is a smooth n^2 -dimensional manifold as an open submanifold.

Example 1.9 (Product smooth manifolds). If $M_1, \dots, M_k$ are smooth manifolds, the product charts $\varphi_1 \times \dots \times \varphi_k$ form a smooth atlas on $M_1 \times \dots \times M_k$, since transition maps split as products of the individual transition maps. This gives the **product smooth structure**, e.g. on the n -torus $\mathbb{T}^n = \mathbb{S}^1 \times \dots \times \mathbb{S}^1$.

1.4 Smooth Maps

The main reason for introducing smooth structures was so we can define smooth functions on manifolds and smooth maps between manifolds.

Definition 1.14 (Smooth functions on manifolds). Suppose M is a smooth n -manifold. $k \in \mathbb{Z}^+$ and $f : M \rightarrow \mathbb{R}^k$ is some function. We say that f is a smooth function if $\forall p \in M$, there exists a smooth chart (U, φ) where $p \in U$ and the composite function $f \circ \varphi^{-1}$ is smooth on the open subset $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$. Recall that a function is *smooth* if it is infinitely continuous and differentiable. We call $f \circ \varphi^{-1}$ the coordinate representation of f and intuitively, it is the function f put into $\mathbb{R}^n$ and we denote this function as $\hat{f}$.

We can generalise the definition of smooth maps between functions very easily now that we know what a smooth map is.

Definition 1.15. Let M and N be smooth manifolds and let $F : M \rightarrow N$ be some map. We say F is a smooth map if for every $p \in M$ there exists smooth charts (U, φ) which contain p and (V, ϕ) containing $F(p)$ such that $F(U) \subseteq V$ and the composite map $\phi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\phi(V)$.

Lemma 1.2 (Every smooth map is continuous). If $F : M \rightarrow N$ is smooth, then F is continuous.

Fix $p \in M$. Since continuity is a local property, it suffices to show F is continuous on some open neighbourhood of p .

By smoothness of F , there exist smooth charts (U, φ) with $p \in U$ and (V, ϕ) with $F(p) \in V$, such that $F(U) \subseteq V$ and

$$\hat{F} := \phi \circ F \circ \varphi^{-1} : \varphi(U) \rightarrow \phi(V)$$

is smooth. Since charts are homeomorphisms, φ and ϕ^{-1} are continuous; and since smooth maps between open subsets of Euclidean space are in particular continuous, $\hat{F}$ is continuous. Then on U we can write

$$F|_U = \phi^{-1} \circ \hat{F} \circ \varphi,$$

a composite of continuous maps

$$U \xrightarrow{\varphi} \varphi(U) \xrightarrow{\hat{F}} \phi(V) \xrightarrow{\phi^{-1}} V \subseteq N,$$

so $F|_U$ is continuous. As U is an open neighbourhood of p , F is continuous at p . Since $p \in M$ was arbitrary, F is continuous. $\square$

Definition 1.16 (Diffeomorphisms). Let M and N be smooth manifolds. A diffeomorphism from M to N is a smooth bijective map $F : M \rightarrow N$ that has a smooth inverse. If such a function exists, we say M and N are diffeomorphic